# MELBOURNE HOUSING PRICE ANALYSIS

Janarrth Srikanth

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# Melbourne Housing Price Analysis

## Project Overview

This Project analyzes over 13,580 properties and the main aim is to determine the key factors that influence the prices of Houses and to help the support stakeholders make data driven decisions based on this analysis.

Dataset Summary

- Rows: 13,580
- Columns: 21
- Key Features: “Price”, “Rooms”, “Type of Property” , “Distance from CBD” and “Year Built”

## 📊 Data Cleaning & Preparation

### 1. Data Loading & Initial Exploration

The Melbourne Housing dataset was imported using the `tidyverse` package and examined using summary and structural functions (`head()`, `glimpse()`, `colSums(is.na())`).

| Suburb | Address | Rooms | Type | Price | Method | SellerG | Date | Distance | Postcode | Bedroom2 | Bathroom | Car | Landsize | BuildingArea | YearBuilt | CouncilArea | Lattitude | Longtitude | Regionname | Propertycount |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| <chr> | <chr> | <dbl> | <chr> | <dbl> | <chr> | <chr> | <chr> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <dbl> | <chr> | <dbl> | <dbl> | <chr> | <dbl> |
| Abbotsford | 85 Turner St | 2 | h | 1480000 | S | Biggin | 3/12/2016 | 2.5 | 3067 | 2 | 1 | 1 | 202 | NA | NA | Yarra | -37.8 | 145. | Northern Metropolitan | 4019 |
| Abbotsford | 25 Bloomburg St | 2 | h | 1035000 | S | Biggin | 4/02/2016 | 2.5 | 3067 | 2 | 1 | 0 | 156 | 79 | 1900 | Yarra | -37.8 | 145. | Northern Metropolitan | 4019 |
| Abbotsford | 5 Charles St | 3 | h | 1465000 | SP | Biggin | 4/03/2017 | 2.5 | 3067 | 3 | 2 | 0 | 134 | 150 | 1900 | Yarra | -37.8 | 145. | Northern Metropolitan | 4019 |
| Abbotsford | 40 Federation La | 3 | h | 850000 | PI | Biggin | 4/03/2017 | 2.5 | 3067 | 3 | 2 | 1 | 94 | NA | NA | Yarra | -37.8 | 145. | Northern Metropolitan | 4019 |
| Abbotsford | 55a Park St | 4 | h | 1600000 | VB | Nelson | 4/06/2016 | 2.5 | 3067 | 3 | 1 | 2 | 120 | 142 | 2014 | Yarra | -37.8 | 145. | Northern Metropolitan | 4019 |
| Abbotsford | 129 Charles St | 2 | h | 941000 | S | Jellis | 7/05/2016 | 2.5 | 3067 | 2 | 1 | 0 | 181 | NA | NA | Yarra | -37.8 | 145. | Northern Metropolitan | 4019 |

**Purpose:**

- Understand dataset dimensions and variable types
- Identify missing values
- Detect potential inconsistencies before transformation

Initial inspection revealed missing values in variables such as `BuildingArea`, `YearBuilt`, and `Landsize`.

---

## 2. Handling Missing Values

### 2.1 Building Area Missing Indicator

```
df <- df %>%
  mutate(BuildingArea_missing = is.na(BuildingArea))
```

Instead of deleting rows with missing building area values, a **binary indicator variable** was created.

**Business Rationale:**

- Missing building area may itself contain information.
- Allows further modelling to detect whether properties with missing building size data behave differently in pricing.

---

## 3. Feature Engineering

Several new variables were created to improve analytical interpretability and business relevance.

---

### 3.1 Year Built Grouping

```
YearBuilt_group = case_when(
  YearBuilt < 1950 ~ 'Pre-1950',
  YearBuilt >= 1950 & YearBuilt < 2000 ~ "1950-1999",
  YearBuilt >= 2000 ~ "2000+",
  TRUE ~ "Unknown"
)
```

Raw construction years were grouped into meaningful historical periods.

**Business Rationale:**

- Buyers think in terms of "older homes vs modern homes", not specific years.
- Enables easier comparison of price premiums across property generations.

---

### 3.2 Date Conversion and Trend Variables

```
Date = dmy(Date)
SaleYear = year(Date)
SaleMonth = month(Date)
```

The sale date was converted into proper date format, and year/month components were extracted.

**Business Rationale:**

- Enables trend analysis over time.
- Supports seasonality detection.
- Makes data compatible with time-series visualisation tools.

---

### 4. Outlier Treatment (Price Capping)

```
price_cap <- quantile(df$Price, 0.99, na.rm = TRUE)
df <- df %>% filter(Price <= price_cap)
```

The top 1% of extreme property prices were removed.

**Business Rationale:**

- Luxury outliers can distort general market trends.
- Focuses analysis on the mainstream residential market.
- Ensures fairer comparison across property attributes.

This maintains interpretability without deleting a large portion of the dataset.

---

## 5. Derived Financial Metrics

### 5.1 Price per Room

```
Price_per_room = Price / Rooms
```

Provides insight into pricing efficiency relative to property size.

---

### 5.2 Price per Square Metre

```
Price_per_sqm = Price / Landsize
```

Normalises price across land size differences.

**Business Rationale:**

- Essential metric for investors.
- Enables regional comparison independent of land size.

---

### 5.3 Property Age at Sale

```
Property_age = SaleYear - YearBuilt
```

Measures the property's age at the time of sale.

**Business Rationale:**

- Age impacts maintenance costs.
- Allows analysis of depreciation or modernity premium.

---

## 6. Location-Based Segmentation

### 6.1 Distance Bucketing

```
Distance_bucket = cut(
  Distance,
  breaks = c(0, 5, 10, 20, 50),
  labels = c("0-5km", "5-10km", "10-20km", "20km+"),
  include.lowest = TRUE
)
```

Distance from CBD was grouped into strategic bands.

**Business Rationale:**

- Buyers evaluate proximity in zones, not raw kilometres.
- Supports urban planning and accessibility analysis.
- Improves dashboard interpretability.

---

## 7. Standardisation of Categorical Variables

### 7.1 Property Type Recoding

```
Type = recode(Type,
              "h" = "House",
              "u" = "Unit",
              "t" = "Townhouse")
```

Converted coded categories into descriptive labels.

**Business Rationale:**

- Improves readability for stakeholders.
- Essential for dashboard presentation.

---

### 7.2 Suburb Name Cleaning

```
Suburb = str_trim(Suburb)
```

Removed leading/trailing spaces in suburb names.

**Business Rationale:**

- Prevents duplicate category issues.
- Ensures accurate grouping and aggregation.

---

# 📈 Exploratory Visual Validation

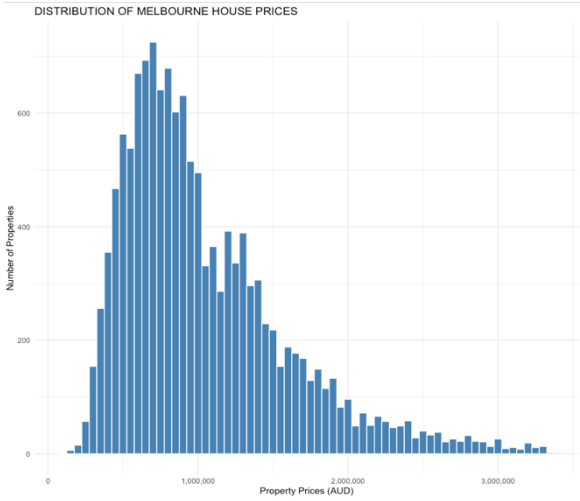
## R-Studio Visuals

Following cleaning and transformation, multiple visualisations were created to validate assumptions and identify pricing patterns:

- Distribution of house prices (histogram)
- Property type vs price (boxplot)
- Distance vs price (scatter + LOESS trend)
- Median price by region (bar chart)
- Median price per square metre by region
- Price by number of rooms (capped at 6)

These visualisations confirmed:

- Positive relationship between property size and price
- Negative relationship between distance and price
- Regional price disparities
- Premium for houses compared to units

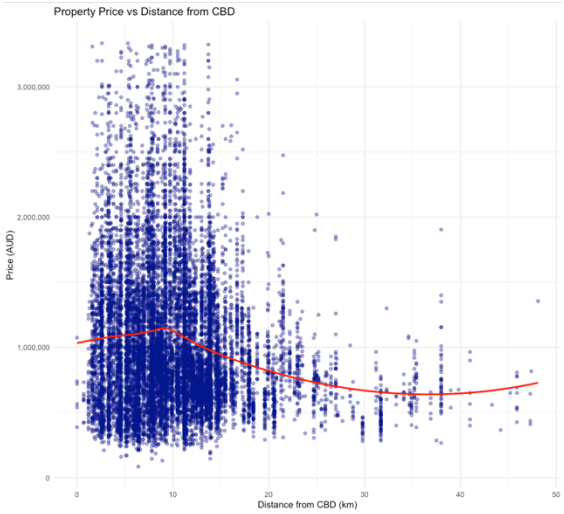



a) Price Distribution Histogram

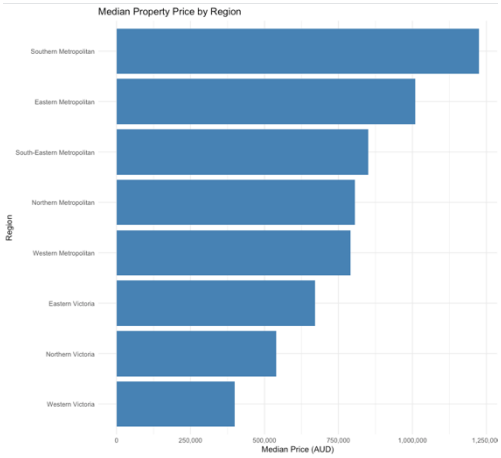
- General Distribution of House Prices across Melbourne
- The Median Price was calculated to be A$900,000.

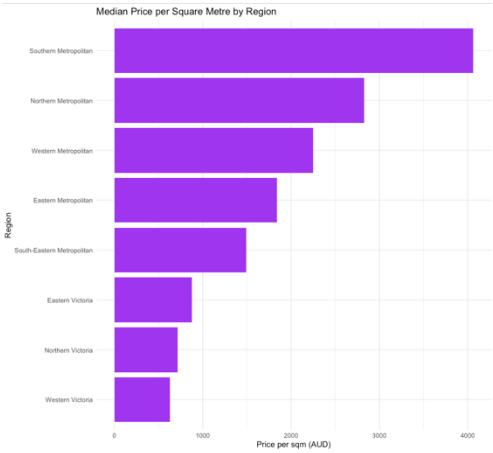
b) Price vs Distance Plot

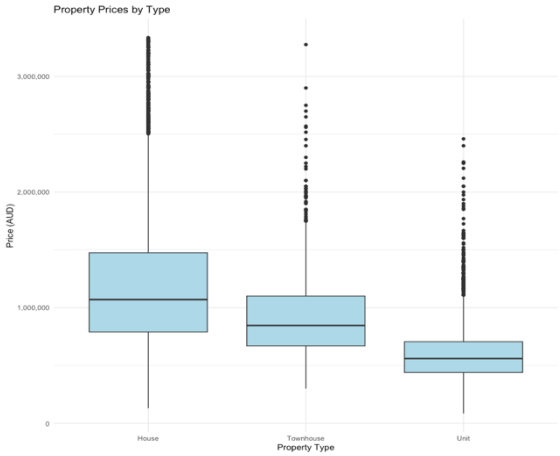
- Initially shows a steady pricing from the Melbourne City Centre but has an inverse relationship following that.
- Prices seem to reduce moving further away from the City Central.




c) Price by Region






d) Price by Property Type

- Box plot of Prices by types such as House, Townhouse and Units.

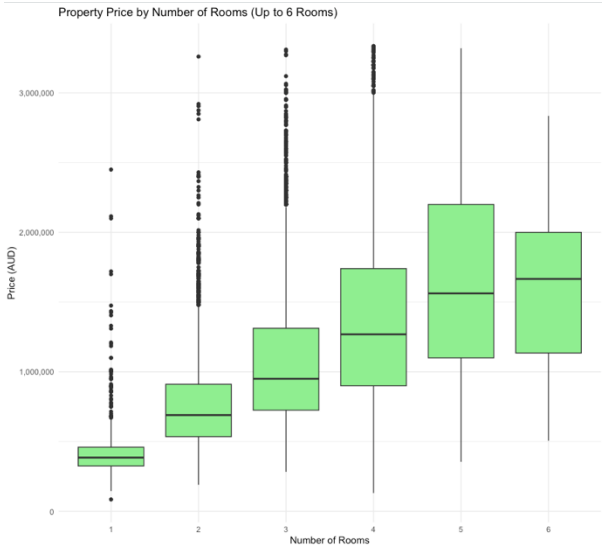
- Individual House usually are more expensive followed by Townhouse and then Units.

e) Price by Rooms

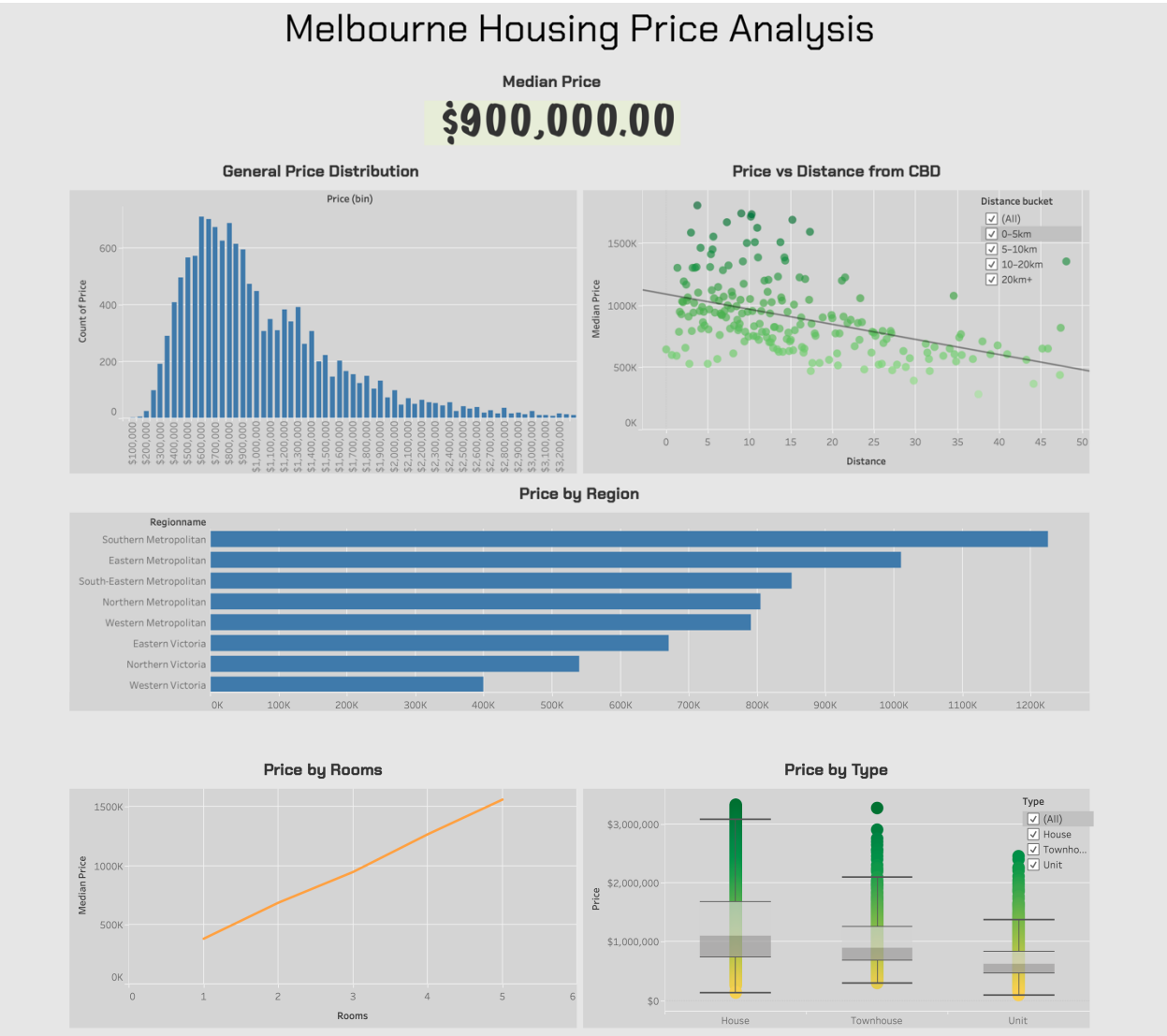
- Boxplot of Prices of houses depending on the number of rooms.

- Houses up to 5 bedrooms have a steady increase in price with each increasing room.

- Houses above 5 rooms are not considered due to lack of sufficient Data.



## Tableau Visuals

Link :
https://public.tableau.com/app/profile/janarrth.srikanth6573/viz/Visualization_f/Dashboard1?publish=yes

And the below is the **Tableau Dashboard** I have created in relation to this: